Younsoo Park

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Education

Pennsylvania State University, University Park, PA — Aug 2021 – Present
B.S. in Computer Science (Math minor), Dean's List — Expected Graduation: May 2027

Experience

Levit Inc. (Shopport AI) - Associate Problem Solver — June 2026 – Aug 2026
- Worked directly with the **CEO** on Shopport, Levit's live iOS/Android AI shopping app, across routing, evaluation, and personalization systems.
- Evolved routing from LLM classification to an embedding-based semantic router (∼**8× faster**) and a hybrid classifier, reaching **86%** V3 accuracy and rescuing **93%** of eligible errors.
- Designed and shipped Wowness, a Mem0-backed cross-category recommender, as sole engineer, and built the production evaluation & backfill platform end-to-end.
- Diagnosed system-boundary production failures and corrected analytics false positives; contributed **730 commits** and **143 merged PRs** (+114K LOC).

Penn State Advanced Vehicle Team (AVT) - Project Manager — Fall 2026 – Present
- Manage planning and deliverables for Penn State's Year 1 EcoCAR Challenge team (Argonne National Laboratory; sponsors GM, MathWorks), building processes later teams inherit.

Undergraduate Research Assistant & Team Lead - Penn State — Oct 2025 – Present
- Researched federated and adversarially robust TinyML for IoT security (manuscript targeting ACM LCTES '26), improving attack recall from 46.7% to 93.9% and F1 from 84.1% to 89.3%.
- Compressed deployment models by 12.3× and reduced ESP32 inference latency by 74.5% across 48 evaluated configurations using pruning, distillation, and INT8 quantization.

SIGNUM (Nittany AI Alliance – PIT-UN Funded) — Aug 2024 – Nov 2025
Co-founded an AI-powered caregiver support platform that received $3,750 in PIT-UN funding.
- Built a HIPAA-aware RAG assistant using AWS Bedrock, FAISS, and spaCy.
- Integrated CMS, NPPES, and Google Places APIs, cutting caregiver research time from 50+ hrs/week to near-instant.

Republic of Korea Air Force — Sept 2022 – June 2024
- Promoted to Squad Leader; led 20+ airmen in high-tempo operations; Commendation Award.

Atom Tech Solutions LTD, Internship — Berkeley, CA — May 2022 – Aug 2022
- Built SQL/JWT authentication, encrypted credential storage, and account automation.

Projects

AI Agent Memory Systems — Jul 2026 – Present
Researched 5 agent memory frameworks (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA approach, under Prof. Kiwan Maeng.

XIIO – LinkedIn for Filmmakers — Apr 2026 – Present
Building a LinkedIn-style platform for filmmakers with creator portfolios, video streaming, and collaboration using Next.js, Firebase, and Cloudflare Stream.

ASME @ Penn State Website — Aug 2024 – Present
Built the ASME Penn State website with React, TypeScript, and Firebase, with real-time Firestore listeners and role-based approval workflows.

Skills

Languages: Python, TypeScript, JavaScript, C++, Java, SQL
Frameworks: React, Next.js, Node.js, NestJS, Flask
AI/ML: PyTorch, TensorFlow, OpenAI API, AWS Bedrock, FAISS, Mem0, scikit-learn, spaCy
Databases & Tools: MongoDB, Redis, Firebase, Kubernetes, Git, Docker, Datadog

Awards & Honors

- 1st Place, ThinkNeuro LLC Project Competition — Apr 2026
- 1st Place, IEEE Student Chapter Battle Bots Competition — Apr 2026
- PIT-UN Funding Award ($3,750) – Nittany AI Challenge — Apr 2025
- 3rd Place, ACM Pathfinder Robotics Challenge — Apr 2025

Activities

Penn State THON™, Technology Captain — Apr 2026 – Present
Lead year-round tech operations and ship features for THON, which raises $10M+ annually.

ACM, Web & Design Director — Apr 2026 – Present

Lead Teaching Assistant, Math 230 (Multivariable Calculus) — Aug 2025 – Present